MELEK BOUABID

Computer Engineering Student — Data Science & Artificial Intelligence
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Profile

Computer Engineering student, specialized in Data Science and Artificial Intelligence at Tek-Up. Experience in Machine Learning, NLP and RAG systems, with projects covering data preparation, model training and deployment.

Technical Skills

- **Programming:** Python, SQL, JavaScript
- **Data Science & ML:** Pandas, NumPy, Scikit-learn, XGBoost, Random Forest, feature engineering
- **Deep Learning & AI:** TensorFlow, PyTorch, NLP, LangChain, RAG, vector databases
- **Data Engineering:** PySpark, PostgreSQL, MySQL
- **Deployment & Tools:** FastAPI, Docker, MLflow, Git
- **Web Development:** React, Django

Education

Computer Engineering Degree — Data Science & AI *2024 – 2027*
Tek-Up, Tunisia — 5th year (in progress)

B.Sc. in Environmental Physics-Chemistry *2019 – 2023*
Higher Institute of Environmental Sciences and Technologies of Borj Cédria, Tunisia

Professional Experience

Data Science / AI Intern — STIET-Philips *2026 (2 months)*

- Contributed to a smart CMMS POC (Laravel 12) for tracking medical installations (MRI and CathLab rooms), structuring operational, technical, financial and documentary data across the installation lifecycle.
- Implemented a conversational assistant based on a documentary RAG architecture (document indexing and chunking, relevant passage retrieval, answer generation via the Gemini API) for administrators and managers.
- Designed decision-support scoring models: equipment risk score (warranty, age, incidents, complaints, maintenance history), team performance score, and statistical prediction of installation duration from archived data.
- Took part in designing an interactive installation-readiness graph (equipment, components, safety elements, documents) with status indicators, and in setting up caching and re-indexing mechanisms.

Mobile Development Intern — Advanced Technologies Software *2025 (1 month)*

- Built a Flutter application to scan Ray-Ban products' barcodes and QR codes and check stock availability.

Projects

SolarVision — Photovoltaic Production Forecasting (France, Tunisia, Spain) *2026*

- **Benchmark** of 12 **time series** forecasting models at 1–7 day horizons (9 cities, 2005–2023, PVGIS, ERA5 and NASA POWER data): persistence, climatology, ARIMA-Fourier, SARIMAX, VAR, Prophet, XGBoost, LSTM, GRU, hybrid ARIMA + LSTM.
- Walk-forward evaluation (RMSE, MAE, MAPE), **Diebold-Mariano** tests with Holm correction, and calibrated 95% **prediction intervals** (GARCH); analyses in both R and Python.
- **MLOps** pipeline: data quality checks, versioned model registry, atomic promotion, GitHub Actions CI; deployed with FastAPI + React and Docker. *Technologies: Python, R, PyTorch, statsmodels.*

CSIS — Criminal Study Intelligent System *2025*

- Social risk detection among young people: **ML** (XGBoost, Random Forest) + **NLP** on text data. Legal assistant on a **RAG** architecture (LangChain, **vector database**) grounded in Tunisian law, presented via Streamlit & Power BI.

Riot Games Analytics Platform *2026*

- Built classification and regression models to predict match outcomes and player rank.
- Data pipeline with the Riot Games API, experiment tracking with MLflow, deployed with FastAPI and React.

Generator Data Crime

- Generated synthetic criminology datasets (JSON/CSV) with realistic statistical correlations, used to test data processing pipelines. *Technologies: React, Vite.*

The Room — Escape Game Booking Website

- Public online booking website with an admin dashboard, real-time notifications and data export. *Technologies: React, Express, PostgreSQL, Docker, SSE.*

E-Bike Rental Platform

- Peer-to-peer electric bike rental marketplace with authentication and online payments. *Technologies: Django REST Framework, React, TypeScript, JWT, Stripe.*

Languages

Arabic: Native | French: Fluent | English: Fluent